Fabric Warehouse DBA

Complete Training Syllabus

Microsoft Fabric Warehouse Database Administration
From Foundations to Expert-Level Administration

15 Topics	80+ Hours	200+ Sub-Topics	Beginner to Expert

April 2025 | Version 2.0 | Enhanced Edition

How to Read This Syllabus

This syllabus combines your original handwritten notes with additional topics required to make it a complete, professional Fabric Warehouse DBA curriculum. Use the colour coding below throughout the document:

* Topics from your original syllabus	+ New topics added to complete the syllabus

GUIDE: All topics are in English. The syllabus is designed for progression — complete topics 1 through 15 in order for the best learning experience.

Table of Contents

1 Introduction to Microsoft Fabric 6 hrs

1.1 Microsoft Fabric Overview (Fabric Workspace Basics)

* **What is Microsoft Fabric — platform overview and value proposition**
* **Capabilities of Fabric — unified analytics, AI, data engineering**
* **Microsoft Fabric Architecture — core platform design**
* **Components of Microsoft Fabric:**
  - Data Factory — data movement and orchestration
  - Analytics — Lakehouse comes under this workload
  - Databases — Warehouse comes under this workload
  - Real-Time Intelligence — streaming and event processing
  - IQ — AI and machine learning workloads
  - Power BI — Reports and visualization
* **Fabric Platform Foundation:**
  - Copilot — AI Assistant embedded across Fabric
  - OneLake — unified storage for all workloads
  - Governance — Security and compliance platform
* **Officially under Data Warehouse workload in Analytics, but operationally managed like a database platform**
* **OneLake Fundamentals and Architecture Overview:**
  - OneLake architecture, storage concepts, and integration with Lakehouse and Warehouse
  - Security, governance, shortcuts, OneCopy concepts, and administration best practices

1.2 Additional Platform Concepts (added)

+ **Fabric vs Azure Synapse Analytics — migration and comparison**
+ **Fabric vs Azure SQL Database — when to use each**
+ **Delta Lake fundamentals — why Parquet + transaction log matters for DBA**
+ **OneLake vs ADLS Gen2 — storage hierarchy and relationship**
+ **Fabric Tenant architecture — tenant, capacity, workspace, item hierarchy**
+ **Fabric API basics — REST API for automation and administration**

2 Microsoft Fabric Licensing & Cost 3 hrs

2.1 Licensing Fundamentals

* Capacity Units (CUs) — understanding the Fabric billing unit
* Trial Capacity — how to get started with free Fabric access
* F-SKU Basics — F2, F4, F8, F16, F32, F64, F128, F256 and above
* Licensing overview — individual user, capacity, and Power BI Pro licenses
* Cost optimization strategies — right-sizing capacity, pause/resume, scheduling

2.2 Advanced Cost Management (added)

+ Burstable capacity and smoothing algorithm — how CU debt works
+ Fabric Capacity Metrics App — monitoring CU consumption by workload and user
+ Throttling behavior — what happens when CU debt exceeds limits
+ Shared vs dedicated capacity — multi-tenancy considerations
+ Cost comparison: Fabric vs Azure Synapse Dedicated SQL Pool
+ Azure Cost Management integration for Fabric billing reporting

3 Workspace in Microsoft Fabric & Power BI 4 hrs

3.1 Workspace Concepts

* Workspace concept and architecture — workspaces as logical containers
* How to create and manage workspaces — naming conventions, settings
* Workspace Roles — Admin, Member, Contributor, Viewer roles and permissions
* Workspace Access Control — permissions and access management
* Deployment Pipelines in Fabric — Dev to Test to Prod promotion approach
* Apply Sensitivity Labels in Fabric and Power BI — Governance and compliance basics

3.2 Advanced Workspace Administration (added)

+ Workspace settings — compute, storage, and network configuration
+ Managed Private Endpoints — securing workspace network connectivity
+ Workspace Git integration — connecting to Azure DevOps or GitHub
+ Workspace item types — understanding all Fabric artifact types
+ Workspace monitoring — usage reporting and audit logs via Admin Portal
+ Multi-workspace architecture patterns — hub-spoke, domain-based design

4 Fabric Warehouse Fundamentals 6 hrs

4.1 Core Warehouse Architecture

* Fabric Warehouse — SQL-based analytics engine overview
* Warehouse Architecture — create a Warehouse in MS Fabric
* Tables in Fabric Data Warehouse — Fact, Dimension, Integration tables, and #Temp tables
* Table Design — data types in Fabric Data Warehouse
* Collation, statistics for tables
* Constraints — Primary Key, Foreign Key, and Unique Key
* Views, Stored Procedures, Functions in Warehouse

4.2 Extended Warehouse Objects & Capabilities (added)

+ Schemas — creating and using schemas for object organization
+ Inline Table-Valued Functions (iTVFs) — creation and usage patterns
+ Cross-database queries — joining Warehouse and Lakehouse tables in T-SQL
+ SQL Analytics Endpoint — read-only T-SQL access to Lakehouse Delta tables
+ Supported vs unsupported T-SQL features in Fabric Warehouse
+ COPY INTO command — syntax, options, and supported file formats
+ External tables and OneLake shortcuts from Warehouse
+ Auto-generated semantic model — Power BI model from Warehouse schema

5 Warehouse Security 6 hrs

5.1 Security Architecture

* Security architecture, permissions, sharing, and encryption
* Microsoft Entra ID Authentication — Azure AD-based login (no SQL auth)
* SQL Granular Permissions — object-level for schema, table, and Warehouse access
* Row-Level Security (RLS) — security predicates in Fabric Warehouse
* Column-Level Security — restricting access to sensitive columns
* SQL Audit Logs — audit and compliance monitoring for Fabric Warehouse
* Dynamic Data Masking — masking PII and sensitive data at query time

5.2 Advanced Security Topics (added)

+ Service Principals and Managed Identities — non-human identity access
+ Azure AD Group-based access — team-level permission management
+ Customer-Managed Encryption Keys (CMK) — bring-your-own-key in Azure Key Vault
+ Private Links and Managed Private Endpoints — network-level security
+ Trusted Workspace Access — secure OneLake access from Azure services
+ Microsoft Purview integration — data classification and sensitivity labeling
+ Security best practices checklist — least-privilege, rotation, audit reviews

6 Fabric Warehouse Capacity & Monitoring 6 hrs

6.1 Your Notes (to be completed)

* Fabric Warehouse Capacity and Monitoring — section started in your notes

NOTE: This section was started in your syllabus but sub-topics were not yet listed. All content below has been added to make it complete.

6.2 Capacity Management (added)

+ Understanding Capacity Units (CUs) and consumption per operation type
+ Burstable capacity — burst thresholds, CU debt, and smoothing window
+ Throttling behavior — when throttling triggers and how to respond
+ Fabric Capacity Metrics App — usage monitoring and breakdown by user/workload
+ Pausing and resuming Fabric capacity — cost management and scheduling
+ Capacity scaling — when and how to scale F-SKU up or down

6.3 Monitoring & Observability (added)

+ Dynamic Management Views (DMVs) — sys.dm_exec_connections, sessions, requests
+ Query Insights schema — queryinsights.exec_requests_history, long_running_queries
+ Monitoring live sessions and killing long-running queries (KILL command)
+ Workload & Concurrency monitoring — concurrent query tracking
+ Billing utilization reporting — query costs and CU attribution
+ Azure Monitor integration — alerts for Fabric service events

7 Data Loading & Ingestion 5 hrs

7.1 Loading Methods (added)

+ COPY INTO command — bulk load from ADLS Gen2, Azure Blob, CSV, Parquet, ORC, JSON
+ INSERT ... SELECT — set-based load from staging or external tables
+ Data Factory Copy Activity — 100+ source connectors into Warehouse
+ Dataflows Gen2 — Power Query M transformations written to Warehouse
+ Spark Notebooks — PySpark reads/writes via OneLake APIs
+ Database Mirroring — near-real-time CDC-based replication from operational DBs

7.2 Loading Patterns & Optimization (added)

+ Staging table pattern — load raw, validate, insert into target atomically
+ Incremental load pattern — delta detection using watermark columns
+ Full load vs incremental load — when to use each strategy
+ Batch size optimization — avoiding too many small Parquet files
+ Parallel loading — partitioning source data for concurrent loads
+ Load performance benchmarks — COPY INTO vs INSERT vs Dataflow
+ Error handling in ETL — logging, retry patterns, failed row capture

8 T-SQL DMV Monitoring & Query Insights 5 hrs

8.1 System DMVs (added)

+ sys.dm_exec_connections — active connections, client details, protocol
+ sys.dm_exec_sessions — session resource usage, status, login info
+ sys.dm_exec_requests — currently running and queued queries
+ sys.dm_tran_active_transactions — open transaction age and state
+ sys.dm_tran_session_transactions — session-to-transaction mapping
+ Role-based DMV access — Admin vs Member vs Contributor permissions

8.2 Query Insights Schema (added)

+ queryinsights.exec_requests_history — completed query records with status, duration, rows
+ queryinsights.exec_sessions_history — historical session-level resource data
+ queryinsights.long_running_queries — aggregated slow query statistics
+ queryinsights.frequently_run_queries — high-frequency query identification
+ Building monitoring dashboards from Query Insights using Power BI
+ Alerting on failed queries and performance SLA breaches
+ Query audit trail — who ran what, when, and from where

9 Transactions, Locking & Concurrency 4 hrs

9.1 Transaction Management (added)

+ ACID properties in Fabric Warehouse — Atomicity, Consistency, Isolation, Durability
+ Snapshot Isolation — the only supported isolation level, behavior explained
+ BEGIN / COMMIT / ROLLBACK TRANSACTION — explicit transaction control
+ Multi-table transactions — spanning multiple tables atomically
+ Implicit vs explicit transactions — auto-commit behavior

9.2 Locking & Concurrency (added)

+ Table-level lock behavior — how INSERT, UPDATE, DELETE acquire locks
+ Write-write conflicts — concurrent UPDATE/DELETE failure scenario and retry pattern
+ INSERTs and immutable Parquet files — why INSERTs rarely conflict
+ Data compaction conflicts — background compaction vs user DML interaction
+ Concurrency limits — max concurrent queries by capacity SKU
+ Workload & Concurrency (4 Hours) — from your notes — detailed scheduling and queuing
+ Error handling patterns — TRY/CATCH, THROW, retry loops in stored procedures

10 Performance Tuning & Optimization
6 hrs

10.1 Query Optimization (added)
+ Statistics management — CREATE, UPDATE, and monitor statistics health
+ Column pruning — SELECT only needed columns for columnar performance
+ Predicate pushdown — filter early, avoid functions on filter columns
+ CTEs vs subqueries — when each improves plan quality
+ Window functions optimization — PARTITION BY, ORDER BY performance
+ APPROX_COUNT_DISTINCT — faster cardinality estimation on large tables

10.2 Table & Schema Design for Performance (added)
+ Star schema design — fact and dimension table modeling for analytics
+ Surrogate integer keys — why INT/BIGINT outperforms NVARCHAR join keys
+ Data type best practices — DATE vs DATETIME, DECIMAL vs FLOAT
+ Avoiding anti-patterns — SELECT *, wide tables, NVARCHAR(MAX) overuse
+ Partitioning strategies — date-based partition elimination

10.3 Data Compaction & Caching (added)
+ Background compaction service — how and when it runs, CU cost
+ Compaction preemption (Oct 2025) — user query priority over compaction
+ Result and data caching — how Fabric caches repeated query results
+ Performance diagnostics scripts — identifying top slow queries, volume trends
+ Autotune Query Optimization — automated plan improvements in Fabric
+ Performance tuning checklist — structured review process for production

11 Disaster Recovery & Business Continuity
5 hrs

11.1 HA and Resilience (added)
+ Azure Availability Zones — automatic zone-level protection, no config needed
+ OneLake geo-redundant storage — how data is replicated at rest
+ Shared responsibility model — what Microsoft manages vs customer manages

11.2 BCDR Configuration & Recovery (added)
+ Enabling BCDR in Fabric Admin Portal — capacity-level toggle and cost implications
+ OneLake asynchronous replication to paired region — RPO implications
+ What is and is NOT replicated — Warehouses, Lakehouses vs Mirrored DBs
+ Restore-in-place — rolling back entire Warehouse to a previous snapshot
+ Cross-region DR strategy — active-active dual-region with dual data loads
+ Post-failover manual steps — roles, permissions, connection strings, reports
+ RTO and RPO planning — defining recovery objectives with business stakeholders
+ DR drill — quarterly testing procedure and validation checklist

12 Source Control, CI/CD & DevOps *5 hrs*

12.1 Git Integration (added)

- + Git integration overview — Azure DevOps Repos and GitHub support
- + Connecting a workspace to Git — step-by-step configuration
- + What gets stored in Git — schema, pipelines, notebooks (not data)
- + Branching strategy — feature, develop, test, main/production branches
- + Fabric Deployment Pipelines — Dev > Test > Production promotion

12.2 CI/CD Automation (added)

- + Azure DevOps Pipelines — automated deployment using Fabric REST API
- + GitHub Actions — workflow YAML for Fabric workspace sync on push
- + Schema migration best practices — idempotent scripts, version numbering
- + CREATE OR ALTER pattern — safe, re-runnable view and procedure scripts
- + Rollback scripts — emergency revert procedures for failed deployments
- + Environment-specific deployment rules — connection string overrides per stage

13 Database Mirroring & Real-Time Sync *4 hrs*

13.1 Fabric Database Mirroring (added)

- + What is Database Mirroring — CDC-based near-real-time replication into OneLake
- + Supported sources — Azure SQL DB, SQL Server, Cosmos DB, PostgreSQL, Snowflake
- + Initial snapshot vs ongoing CDC replication — phase-by-phase explanation
- + Setting up mirroring from Azure SQL Database — enable CDC, configure in Fabric
- + Setting up mirroring from on-premises SQL Server — Data Gateway requirement

13.2 Monitoring & Limitations (added)

- + Monitoring mirror status — replication lag, error detection
- + Mirror latency factors — network, transaction size, compaction
- + Mirroring limitations — primary key requirement, TRUNCATE not captured
- + Mirroring vs ETL pipelines — when to use each pattern
- + Mirror recovery post-DR — recreating mirrors in secondary region

14 Integration with External Services & Tools *5 hrs*

14.1 Microsoft Services Integration (added)

- + Power BI — DirectLake, DirectQuery, and Import mode connectivity
- + Data Factory Pipelines — Copy Activity, Script Activity, ForEach patterns
- + Dataflows Gen2 — Power Query transformations into Warehouse tables
- + Lakehouse — cross-database T-SQL queries across Warehouse and Lakehouse
- + Azure Data Lake Storage Gen2 — COPY INTO from ADLS file storage
- + OneLake Shortcuts — referencing ADLS, S3, GCS without data movement
- + Microsoft Purview — data catalog, lineage, classification, governance

14.2 Third-Party Tools & Connectivity (added)

+ **SSMS & Azure Data Studio — connection string, SQL Endpoint, auth method**
+ **dbt (data build tool) — Fabric Warehouse adapter for SQL transformations**
+ **Python (pyodbc / SQLAlchemy) — programmatic Warehouse access**
+ **Apache Airflow — pipeline orchestration via REST API or SQL operator**
+ **Fivetran / Informatica — SaaS and enterprise ETL connectors**
+ **Tableau, Qlik, Excel — BI tool connectivity via ODBC/JDBC**

15 Fabric Administration & Governance 5 hrs

15.1 Fabric Admin Portal (added)

+ Admin Portal overview — tenant settings, capacity management, governance controls
+ Tenant-level settings — enable/disable features, data residency controls
+ Capacity administration — assign workspaces to capacities, monitor utilization
+ Activity log — audit trail of all admin and user actions across the tenant
+ Usage metrics — workspace-level and tenant-level usage reporting

15.2 Governance & Compliance (added)

+ Microsoft Purview data governance — catalog, glossary, lineage, classification
+ Sensitivity labels — Microsoft Information Protection labels on Warehouse objects
+ Data residency and sovereignty — region selection and compliance requirements
+ GDPR, HIPAA, SOC2 considerations for Fabric Warehouse
+ Domain and data mesh architecture — organizing workspaces by business domain
+ Fabric certification path — DP-600 (Fabric Analytics Engineer) exam preparation

15 Topics	75+ Hours	200+ Sub-Topics	Beginner to Expert

This syllabus is designed as a complete end-to-end curriculum for Microsoft Fabric Warehouse Database Administration. Topics 1-5 cover your original notes. Topics 6-15 add all the missing DBA domains required for production-level expertise.